BST of Taiwan College Departments

Problem Description

You are given a sequence of commands and Taiwan college department records. Maintain a Binary Search Tree (BST) whose key is **School Name**. Each node stores one department record.

Here are the tree rules:

1. When inserting a record, place a smaller School Name in the left subtree and an equal or larger School Name in the right subtree.
2. Records with the same School Name are stored as separate nodes. A search or deletion applies to **all matching records** currently in the tree.
3. When a deleted node must be replaced, use its inorder successor. The height of an empty tree is 0; the height of a leaf is 1.

Input Format

The input consists of command lines and their associated data. Each command line contains exactly one integer from 0 to 3:

- **0** — **Exit:** terminate the program.
- **1** — **Insert:** read one or more department records until the next command line, and insert them into the existing BST.
- **2** — **Search:** read one School Name from the next line.
- **3** — **Delete:** read one School Name from the next line.

Each department record is one line containing the following **11 TAB-separated columns**, in order. Store and print the seven fields shown in bold.

0	School Code	6	**Number of Students**
1	**School Name**	7	**Number of Teachers**
2	Department Code	8	**Graduates**
3	**Department Name**	9	County/City Name
4	**Day/Continuing**	10	System Type
5	**Education Level**		

Spaces may occur within a field. A record starts with a four-character School Code, such as `0001`; a command line contains only `0`, `1`, `2`, or `3`. Do not treat the next command as a record.

Output Format

Print the output for each command as follows:

- **Command 1:** print `Tree height = h`, where `h` is the resulting height.
- **Command 2:** print `Search results:`, followed by all matching records.
- **Command 3:** print `Delete records:`, followed by all records being removed, then print `Tree height = h` for the remaining tree.
- If no matching record exists, print `There is no match!`. For an empty BST, the empty-tree message is `Tree is empty!`.
- **Command 0:** terminate without additional output.

Number records from 1 within each result set, using `[1]`, `[2]`, and so on. Print these seven fields in this order: School Name, Department Name, Day/Continuing, Education Level, Number of Students, Number of Teachers, and Graduates.

Constraints

- Command 0 occurs exactly once, on the final input line.
- Each department record contains exactly 11 TAB-separated columns.
- School Name and Department Name may contain Chinese characters and punctuation.
- The BST persists across multiple insertion batches.
- Duplicate School Names are allowed and must be returned together.

Example Test Case

The sample uses the same three records as the original statement. Each record occupies one input line. Separators between input fields are TAB characters; the displayed spacing is only for readability.

Sample Input 1

```
1
0002	國立清華大學	520114	資訊工程學系	D 日	B 學士	565	43	117	18 新竹市	1 一般
0003	國立臺灣大學	520114	資訊工程學系	D 日	B 學士	520	30	104	30 臺北市	1 一般
0005	國立成功大學	520114	資訊工程學系	D 日	B 學士	461	28	85	21 臺南市	1 一般
2
國立成功大學
3
國立臺灣大學
2
中原大學
0
```

Sample Output 1

```
Tree height = 2
Search results:
[1]	國立成功大學	資訊工程學系	D 日	B 學士	461	28	85
Delete records:
[1]	國立臺灣大學	資訊工程學系	D 日	B 學士	520	30	104
Tree height = 2
Search results:
There is no match!
```

Explanation of Sample 1

The three insertions produce a tree of height 2. Searching for 國立成功大學 returns its department record. Deleting 國立臺灣大學 removes its record and leaves a tree of height 2. The final search for 中原大學 has no match.